Research Methodology in Computer Science

UNIT I WORKSHEET

Introduction to Research Problem



Course: CSEG3060 | Unit: I | Lecture Allocation: 9 Hours
Designed as an active-learning revision and practice worksheet.

Name: _____ **Roll No.:** _____

Section: _____ **Date:** _____

How to use this worksheet

This worksheet is meant to move you from *"I know the definitions"* to *"I can formulate and defend a research problem."* Work through the activities in order. Use your own Computer Science topic whenever a question asks you to design or formulate something.

Unit I Coverage Checklist

- ☐ Meaning and significance of research
- ☐ Objectives of research
- ☐ Key elements of research
- ☐ Sources of research problems
- ☐ Identification of research problems
- ☐ Good research problem criteria
- ☐ Common selection errors
- ☐ Avoiding selection errors
- ☐ Scope and boundaries
- ☐ Research objectives
- ☐ SMART objectives
- ☐ Research approaches
- ☐ Theoretical vs. empirical research
- ☐ Data types and sources
- ☐ Data collection tools
- ☐ Data collection guidelines
- ☐ Data analysis
- ☐ Interpretation of results
- ☐ Research instrumentation
- ☐ Calibration and repeatability
- ☐ Integrated case-based application

1 Part A — Research: What, Why and How?

A1. Research in one sentence

Research is a **systematic and scientific investigation** used to discover new facts, verify existing knowledge, analyze relationships, explain phenomena, and answer questions using evidence.
Key elements from class:

Problem Identification → Data Collection & Analysis → Hypothesis Formation → Result Interpretation

Activity 1: Build the research chain

Put the following in a sensible order:

Result Interpretation	Problem Identification	Data Collection/Analysis	Hypothesis Formation
4	1	2	3

Why is this order useful in a CS research project? **[2 marks]**

Order: 1. Problem Identification, 2. Data Collection/Analysis, 3. Hypothesis Formation, 4. Result Interpretation. Each step builds on the previous one: a clear problem decides what data is needed, the analysed data supports or rejects the hypothesis, and the interpretation answers the problem. Following this order keeps a CS project systematic, focused and repeatable, and avoids building or testing something before it is clear what question is being answered.

Activity 2: Why does research matter?

Give one CS example for each:

Technology/knowledge advancement	Transformer language models (e.g. BERT, GPT) advanced natural-language understanding.
Industrial innovation	Recommendation algorithms at Netflix/Amazon improve user engagement and sales.
Academic growth	Benchmark datasets such as ImageNet enable reproducible ML research and papers.
Policy or societal decision-making	Studies on algorithmic fairness and privacy shaped data-protection rules (e.g. GDPR).
Computer Science example (AI/algorithms/HCI/etc.)	AI: federated learning for privacy-preserving training. HCI: usability testing of apps. Algorithms: faster sorting/searching.

Activity 3: Research objectives

Complete the table.

Objective of research	Your CS example
Find answers using scientific methods	Run controlled experiments to find which sorting algorithm is fastest on large datasets.
Explain phenomena / establish relationships	Explain how dataset size or model depth affects accuracy and training time (a relationship).
Test or validate an existing theory	Test the claim that merge sort runs in O(n log n) by timing it on growing inputs.

Think like a researcher

A student says: *"I want to build an AI app because AI is trending."* Is this already a research problem? Explain what is missing.

No. 'AI is trending' is only a topic and a motivation, not a research problem. Missing: a specific problem or research gap, a research question, objectives, scope/boundaries, a method with data and evaluation plan, and novelty. Building an app is a development task unless it answers a clearly stated question.

2 Part B — Types and Approaches of Research

Research approaches covered in Unit I

Exploratory
discover patterns/ideas

Descriptive
describe facts/characteristics

Analytical
analyze information

Theoretical
models, logic, proofs

Empirical
observation/experiment

The class material also introduced **fundamental** and **applied** research as broad research types. Choose the approach according to the research question and the kind of evidence required.

Activity 4: Match the approach

Write E (Exploratory), D (Descriptive), A (Analytical), T (Theoretical), or EM (Empirical).

1. Investigating an emerging problem where very little is known: E
2. Measuring average response time of five search algorithms: EM
3. Developing a formal model for a new computational system: T
4. Summarizing user characteristics from a usability survey: D
5. Comparing existing datasets and identifying relationships among variables: A

Activity 5: Theoretical or empirical?

For each scenario, decide the dominant approach and justify it in one sentence.

Scenario	Type	Why?
Prove a property of an algorithm mathematically	Theoretical	Proof relies on logic and formal reasoning, not on data or experiments.
Benchmark an intrusion-detection model on a dataset	Empirical	Uses experiments and measured results (accuracy, detection rate) on a dataset.
Develop a formal model of a distributed protocol	Theoretical	Builds an abstract formal model using mathematics and logic.
Interview developers to discover pain-points in an emerging tool	Empirical (exploratory)	Collects observations by interviews to discover new patterns in an emerging area.

3 Part C — Finding and Formulating a Research Problem

C1. Where do research problems come from?

Important sources highlighted in class include:

- Literature reviews and research gaps.
- Industrial problems and real-world needs.
- Emerging trends and technologies.
- Academic coursework and past projects.
- Surveys of recent publications.
- Patent and research databases.
- Industry white papers.
- Competitive research proposals.

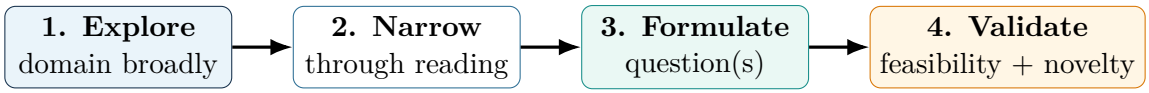
Activity 6: Research-problem detective

Pick a CS domain: Network security: machine-learning-based intrusion detection

List **four possible sources** from which a research problem in this domain could emerge:

1. Literature review: survey papers show ML-based IDS often miss new (zero-day) attacks, a clear research gap.
2. Industrial problems: industry white papers and security reports show rising ransomware and DDoS incidents.
3. Emerging trends: IoT and 5G networks create new attack surfaces and large volumes of encrypted traffic.
4. Patent and research databases (IEEE Xplore, Google Scholar) reveal unsolved problems and available datasets.

C2. Identification workflow



Activity 7: Apply the four-step workflow

Topic/domain: ML-based network intrusion detection for IoT networks

1. Explore broadly: What is the larger area?
Cybersecurity, then network security, then intrusion detection systems (IDS).

2. Narrow through reading: What specific sub-area or pain-point interests you?
Detecting unseen (zero-day) attacks in IoT network traffic with a low false-alarm rate.

3. Formulate question(s): Write one research question.
How accurately can a lightweight ML classifier detect unseen attacks in IoT traffic compared with a signature-based IDS?

4. Validate: What would you check for feasibility and novelty?
Feasibility: public datasets (UNSW-NB15, CIC-IDS2017), free tools, available time and skills. Novelty: literature search (Google Scholar, IEEE) to confirm the gap is not already solved.

Quick Pair Challenge: Problem or Topic?

Classify each statement as **Topic**, **Weak Problem**, or **Research Problem**.

1. "Artificial Intelligence in Smart Cities." Topic
2. "AI is interesting." Weak Problem
3. "How can reinforcement learning optimize traffic signals in smart cities?" Research Problem
4. "Cybersecurity." Topic
5. Your own statement: How can ML reduce false alarms in IoT intrusion detection?

C3. Characteristics of a good research problem

A strong problem should be **clear and concise, researchable and feasible, relevant and original**, and should offer scope for **data collection and analysis**. The syllabus also emphasizes essential attributes and avoiding errors in selection. **[5 marks]**

Criterion	Evidence that YOUR problem satisfies it
Clear and concise	States one focus (IoT intrusion detection), one method (lightweight ML) and one outcome (detecting unseen attacks) in a single question.
Researchable	Can be investigated through experiments; the answer is measurable with accuracy, recall and false-alarm rate.
Feasible	Needs only public datasets (UNSW-NB15), free tools (Python, scikit-learn) and a standard laptop/Colab within one semester.
Relevant	IoT attacks are increasing; better detection protects users and organisations and is an active research area.
Original / non-trivial	Most work targets known attacks; detecting unseen attacks at low computing cost is a gap, so the problem is non-trivial.
Allows data collection and analysis	Labelled traffic records and captured packets (Wireshark) allow measurement, comparison with baselines and statistical analysis.

Problem used for all 'your own' answers: How can a lightweight ML model detect unseen attacks in IoT network traffic?

4 Part D — What Can Go Wrong? Scope and Boundaries

Common errors in selecting a research problem

1. Ambiguity / vagueness
The problem does not clearly state what is being investigated.
2. Too broad
The study becomes unrealistic or unmanageable.
3. Too narrow
There may be insufficient room for meaningful investigation.
4. Lack of relevance
The problem has weak academic, practical, or societal value.
5. Lack of feasibility
Resources, data, skills, time, or infrastructure are unavailable.
6. Ignoring prior art
Existing research may already answer the question.

Activity 8: Error clinic

For each weak problem, identify the main error and rewrite it.

1. "Improve AI everywhere."
   Error: Too broad and vague (no clear focus).
   Improved version: Improve the accuracy of a CNN image classifier for detecting plant-leaf diseases using the PlantVillage dataset.

2. "Study all cybersecurity threats affecting every organization worldwide."
   Error: Too broad (unmanageable scope).
   Improved version: Analyse phishing attacks on small e-commerce websites during 2024-25 using public threat-report data.

3. "Build a system that detects every possible disease with 100% accuracy in two days."
   Error: Lack of feasibility (unrealistic accuracy, scope and time).
   Improved version: Develop and evaluate a CNN that detects pneumonia from chest X-rays on a public dataset, aiming for at least 90% accuracy in one semester.

How to avoid selection errors

- Perform a literature review.
- Consult experts or guides.
- Conduct pilot studies where appropriate.
- Align the problem with objectives and available resources.

D2. Scope = the boundary of your research

Scope defines the **limits and boundaries** of the study and prevents deviation from the objectives. Typical boundaries include:

Time frame | **Geographic** location | **Target population** or dataset | **Tools /** technologies

Activity 9: Draw the boundary

Research problem: *"How can machine learning improve network intrusion detection?"*

Fill in realistic boundaries:

Time frame	Dataset captured 2015-2019; experiments completed in one semester (about 12 weeks).
Geographic / deployment context	Simulated enterprise/campus LAN and IoT test network; no real-world deployment.
Target dataset / population	Public benchmark traffic records: NSL-KDD / CIC-IDS2017 / UNSW-NB15.
Tools / algorithms	Python, scikit-learn, TensorFlow; Random Forest, SVM and CNN; Wireshark for capture.
What is explicitly OUT of scope?	Live production deployment, intrusion prevention (blocking), hardware/firmware attacks, and cloud-scale or global networks.

Activity 10: Scope test

A student's project is titled *"AI for Smart Cities."* Give three reasons why this scope is likely too broad, then convert it into a narrower study.

Why it is too broad: (1) 'AI' covers many techniques (machine learning, vision, NLP, optimisation) and 'Smart Cities' covers many domains (traffic, waste, energy, water, safety). (2) It names no specific problem, population, dataset or boundary, so it cannot be completed in the available time. (3) It would need huge data and infrastructure, existing work is vast, and no clear gap or measurable objective can be stated.

Narrower study: Using reinforcement learning to optimise traffic-signal timing at a single urban four-way intersection, to reduce the average vehicle waiting time by at least 15% compared with fixed-time signals, evaluated in the SUMO simulator over 12 weeks.

5 Part E — Research Objectives and SMART Thinking

E1. Why objectives matter

Research objectives:

- set the direction and purpose of research;
- help define scope and methodology; and
- provide focus and coherence.

A good objective is not just an activity such as "study AI." It states **what will be achieved or evaluated**.



Activity 11: SMART objective laboratory

Research topic/problem: ML-based intrusion detection for IoT networks

Write one objective, then audit it.

Specific	What exactly will be done? Train and compare Random Forest and CNN classifiers to detect unseen attacks in IoT network traffic.
Measurable	What variable, metric, or evidence will show achievement? Detection accuracy, recall and false-alarm rate; target at least 95% accuracy and under 3% false alarms.
Achievable	Why is it realistic with available resources? Uses the public UNSW-NB15 dataset, free Python tools and a standard laptop/Colab within student skills.
Relevant	How does it address the research problem? Directly addresses the problem of detecting new attacks in IoT traffic with machine learning.
Time-bound	By when / over what period? Completed within 12 weeks: data in weeks 1-3, models in weeks 4-9, analysis in weeks 10-12.

Final SMART objective:
Within 12 weeks, train and compare Random Forest and CNN models on the UNSW-NB15 dataset to detect unseen IoT attacks, achieving at least 95% accuracy with under 3% false-alarm rate.

Peer Review — 60-second test

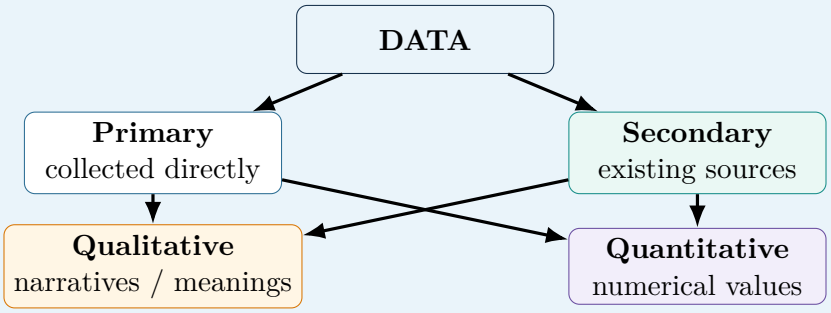
Exchange your objective with a partner. Your partner must answer:

1. Can I identify exactly what is being investigated?
2. Can I identify how success will be measured?
3. Is the target realistic?
4. Does it directly connect to the problem?
5. Is a time boundary visible?

Partner's one improvement suggestion: Add a baseline model and state the train/test split so results can be verified.

6 Part F — Data Fundamentals

F1. Two useful ways to classify data



Examples discussed in class include primary interviews, sensor readings, and usage logs; secondary GitHub/Kaggle datasets, IEEE articles, and government data.

Activity 12: Classify the data

Mark each as Primary/Secondary and Qualitative/Quantitative where applicable.

Data source	Primary/Secondary	Qualitative/Quantitative
Interview transcripts	Primary	Qualitative
Sensor temperature readings	Primary	Quantitative
Kaggle dataset downloaded for analysis	Secondary	Quantitative
System CPU utilization logs collected during your experiment	Primary	Quantitative
User comments about usability	Primary	Qualitative
IEEE article used as an existing source	Secondary	Qualitative

F2. Choosing data

For your own research problem:

1. What data do you need?
   Labelled network-traffic records (normal and attack flows) with features such as duration, packets, bytes and protocol.

2. Why is this data appropriate?
   It has known attack labels and realistic traffic, so models can be trained, tested and compared objectively.

3. Primary, secondary, or a combination?
   A combination: secondary benchmark data (UNSW-NB15) plus primary traffic captured in a lab with Wireshark.

4. Qualitative, quantitative, or a combination?
   Mostly quantitative (flow features, accuracy, recall); qualitative only for attack-type labels.

7 Part G — Data Collection Tools and Guidelines

G1. Match the tool to the evidence

The class material highlighted:

- **Questionnaires / online forms** — structured responses from participants.
- **Observation logs / experiments** — record observed behavior or experimental events.
- **System monitoring tools** — capture system/network performance.
- **Web scraping** — collect information from web sources using appropriate tools.

Examples shown in the class material include Google Forms, Typeform, Wireshark, PerfMon, BeautifulSoup, and Selenium.

Activity 13: Tool selection game

You need to collect the following evidence. Select the most suitable tool and justify.

Evidence required	Tool / method	Why?
Students' satisfaction with an app	Questionnaire / Google Forms	Collects structured responses from many students that are easy to compare.
Network packet behavior during an experiment	Wireshark	Captures and inspects the packets passing through the network.
CPU/memory usage of a program	PerfMon (system monitoring)	Records CPU and memory counters over time while the program runs.
Behavior of users interacting with a prototype	Observation log / experiment log	Records what users do with the prototype, step by step.
Public web data needed for a study	Web scraping (BeautifulSoup / Selenium)	Extracts public web data automatically and at scale.

Activity 14: Design a mini data-collection instrument

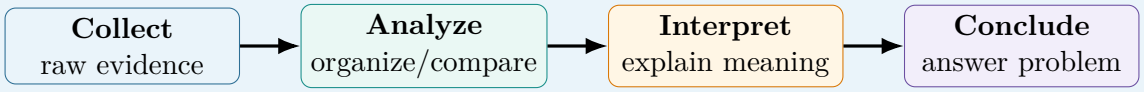
Choose one: ☐questionnaire ☐observation log ☐experiment log ☒system-monitoring plan.

Research problem: Detecting unseen attacks in IoT network traffic using ML

What will be measured/recorded?	Packets/flows per second, bytes, protocol, source/destination, attack label and CPU use during each run.
Who/what is the source?	Lab IoT test network with a simulated attack machine; UNSW-NB15 as secondary reference data.
When/how often?	Continuous capture for 30 minutes per run; 10 runs per scenario (normal and attack) over 2 weeks.
What tool will be used?	Wireshark for packet capture and PerfMon (or top) for CPU and memory monitoring.
How will consistency be maintained?	Same hardware and settings, fixed scripts and capture duration, repeated runs, clock/sensor checks and a configuration log.

8 Part H — From Data to Meaning: Analysis and Interpretation

H1. Do not confuse these stages



Analysis asks: "What patterns, differences, relationships, or measurements are in the data?"
Interpretation asks: "What do those findings mean for my research problem and objectives?"

Activity 15: Read the result, then interpret it

Suppose an experiment gives:

Model	Accuracy	Inference time (ms)
A	92%	20
B	95%	70
C	94%	25

1. Which model has the highest accuracy? Model B (95%)

2. Which model is fastest? Model A (20 ms)

3. If the system is real-time and accuracy must be at least 93%, which model would you investigate first? Why?
   Model C. It meets the accuracy requirement (94% is at least 93%) and is fast (25 ms). A is fast but only 92% accurate; B is accurate but slow (70 ms), so C best fits a real-time system.

4. Is "B is best" a valid conclusion without considering the research objective? Explain.
   No. 'Best' depends on the objective and constraints. B has the highest accuracy but is 3.5 times slower than A, so for a real-time objective it may not be best. A conclusion must be linked to the research objective.

Activity 16: Interpretation vs. description

Complete each sentence with either **description** or **interpretation**.

1. "Model B achieved 95% accuracy." description

2. "The higher accuracy may be useful when accuracy is the primary objective, but its latency may limit real-time deployment." interpretation

3. "The system processed 10,000 requests." description

4. "The observed throughput suggests the system can support the target workload." interpretation

Researcher's caution

A graph can show a pattern, but a pattern is not automatically an explanation. Write one question you would ask before claiming *why* a result occurred.
Could another factor (data imbalance, hardware, random seed) explain this pattern, and can the result be repeated?

9 Part I — Research Instrumentation

I1. Instrumentation in computing research

Research instrumentation refers to the **tools, hardware, and software used for data collection and analysis**. Class examples include:

- sensors;
- simulators;
- benchmarking suites;
- custom tools for robotics, networks, and AI.

Two important quality ideas emphasized in the class material are **calibration** and **repeatability**.

Activity 17: Instrument selection matrix

For your chosen research problem, identify one instrument at each level.

Level	Possible instrument	What evidence will it produce?
Hardware / sensor	Network interface / TAP on a test router	Raw packet traffic from IoT devices
Software / simulator	Mininet / NS-3 network simulator	Simulated normal and attack traffic
Benchmark / measurement tool	UNSW-NB15 dataset with scikit-learn metrics	Accuracy, precision, recall, F1, false-alarm rate
Custom tool (if needed)	Python script extracting flow features and logging runs	Feature tables and run logs for analysis

Activity 18: Calibration and repeatability

Why should a sensor, measurement setup, benchmark, or custom instrument be calibrated?
Calibration compares the instrument with a known reference so readings are accurate and free from drift or bias; without it, measurements and conclusions cannot be trusted.

What could happen if the same experiment cannot be repeated consistently?
Results cannot be verified or reproduced by others, comparisons become unreliable, findings may be due to chance or setup errors, and the research loses credibility.

10 Part J — The Research Problem Builder

Your 10-minute research design

Choose a real Computer Science problem you care about. Complete every box.

Step 1 — Domain

Computer networks / cybersecurity: machine-learning-based intrusion detection.

Step 2 — Real-world pain-point

IoT networks face new (zero-day) attacks that signature-based IDS cannot detect, causing data loss and downtime.

Step 3 — Research problem statement

Existing ML-based IDS work well on known attacks but poorly on unseen attacks in resource-limited IoT networks.

Step 4 — Research question

How can a lightweight ML model detect unseen attacks in IoT network traffic better than a signature-based IDS?

Step 5 — Why is it worth investigating?

Better detection protects users and organisations, reduces losses and fills a gap in unseen-attack detection for low-resource IoT.

Step 6 — Research approach

☐Exploratory ☐Descriptive ☐Analytical ☐Theoretical ☒Empirical ☐Fundamental ☒Applied

Step 7 — Scope boundaries

Time: 12 weeks Dataset/population: UNSW-NB15 (IoT traffic)

Tools/technology: Python, scikit-learn, TensorFlow, Wireshark Out of scope: Live deployment, prevention

Step 8 — Three SMART objectives

O1: Collect and preprocess the UNSW-NB15 dataset into a clean labelled training set within 3 weeks.

O2: Train and compare Random Forest and CNN models on unseen attacks, reaching at least 95% accuracy within 9 weeks.

O3: Evaluate false-alarm rate (under 3%) and inference time against a signature-based IDS baseline by week 12.

Step 9 — Data plan

Data type(s): Quantitative (traffic flows) Source(s): UNSW-NB15; lab captures

Collection tool(s): Wireshark, Python scripts Instrumentation: Mininet, scikit-learn

Step 10 — Analysis and interpretation

What will you measure/analyze? How will the findings answer the research question?

Measure accuracy, precision, recall, F1 and false-alarm rate of each model. Analysis compares the models with the baseline; interpretation links these results to the objectives to show whether ML detects unseen IoT attacks better.

11 Part K — Case Study: AI in Smart Cities

Case Study — Refine the vague idea

A student wants to research:

"Artificial Intelligence in Smart Cities."

Your task is to transform this broad idea into a defensible Unit I research design.

K1. Narrow the problem

Broad area: **AI + Smart Cities**

Choose one specific application: ☒traffic ☐waste ☐energy ☐water ☐safety ☐other: _____

Specific pain-point:
Fixed-timer traffic signals cause long vehicle waiting times, congestion and fuel waste at busy urban junctions.

Refined research problem:
How can reinforcement-learning-based signal control reduce the average vehicle waiting time at a single urban intersection compared with fixed-time signals, using SUMO simulation?

K2. Define the approach

Dominant approach: Empirical (applied)

Why is it theoretical, empirical, exploratory, descriptive, analytical, or a combination?
Empirical, because the RL controller is tested through simulation experiments and measured (waiting time, queue length). It is applied research on a real traffic problem, with an exploratory literature review to find the gap.

K3. Three SMART objectives

1. Within 3 weeks, build a SUMO model of one four-way junction using open traffic-flow data, with a fixed-time baseline.

2. Within 8 weeks, train a Q-learning/DQN signal controller that cuts average waiting time by at least 15% versus fixed-time control.

3. By week 12, evaluate waiting time, queue length and throughput over 30 simulation runs and report the results.

K4. Boundaries

Timeframe	12 weeks (one semester); one simulated peak hour of traffic per run.
Target dataset / population	One four-way intersection; open traffic-count / vehicle-flow data.
Geographic/deployment context	Single urban junction in simulation only (no real-world deployment).
Tools/technologies	SUMO simulator, Python (TraCI), Stable-Baselines3 / TensorFlow.
Out of scope	Multiple intersections, pedestrians, emergency vehicles, hardware deployment and other smart-city domains (waste, energy).

12 Part L — Case-Based Thinking

Case 1: "Build a better chatbot"

A student proposes: *"I will make a chatbot that is better than ChatGPT."*
Identify at least **four** problems with this research proposal at the Unit I level.

1. Too broad and vague: 'better' is not defined and there is no metric.
2. Not feasible: ChatGPT needs huge data, compute and teams that a student cannot access.
3. No research question, objectives, scope or method; it is a build task, not research.
4. Ignores prior art and has no data or evaluation plan, so success cannot be measured.

Rewrite it as a focused research problem:
How does fine-tuning a small language model on university FAQ data affect answer accuracy and response time compared with a rule-based chatbot, measured on 200 test questions within 10 weeks?

Case 2: Sensor-based smart agriculture

A team wants to predict irrigation needs using soil-moisture sensors.
Identify:

1. One possible research problem. Predicting irrigation need from soil-moisture readings using machine learning.
2. Two data sources. Field soil-moisture sensors; public weather/rainfall data.
3. One primary data source. Readings from soil-moisture sensors in the field.
4. One secondary data source. Public weather/rainfall dataset (e.g. from the meteorological department).
5. One data-collection tool. Arduino / ESP32 data logger.
6. One research instrument. Capacitive soil-moisture sensor.
7. One measurable objective. Predict irrigation need with at least 90% accuracy within 8 weeks.
8. One scope boundary. One crop (wheat) on a single one-acre field for one season.

Case 3: The "perfect accuracy" trap

A researcher claims: *"My model is successful because it gives 99.9% accuracy."*
Write three questions you would ask before accepting the claim:

1. Is the 99.9% measured on unseen test data, or on training data (overfitting / data leakage)?
2. Is the data imbalanced? What are the precision, recall and F1, not just accuracy?
3. Can the result be repeated on other datasets and runs and compared with baseline models?

13 Part M — Unit I Rapid-Fire Quiz

Choose the best answer.

1. A research problem should primarily be:

   (a) fashionable
   (c) extremely broad
   (b) clear and researchable
   (d) impossible to test

2. Which is a source of research problems?

   (a) literature gaps
   (c) avoiding prior work
   (b) random guessing only
   (d) unrelated facts

3. "Measure the average latency of Algorithm X on Dataset Y within four weeks" is closest to:

   (a) vague topic
   (c) literature review
   (b) SMART objective
   (d) research instrument

4. Existing data from Kaggle is generally:

   (a) primary
   (c) always qualitative
   (b) secondary
   (d) always experimental

5. "User comments about usability" are primarily:

   (a) qualitative
   (c) hardware
   (b) quantitative
   (d) benchmark

6. Which is most directly related to system monitoring?

   (a) PerfMon
   (c) patent claim
   (b) title page
   (d) abstract

7. A study based mainly on observation and experiments is:

   (a) empirical
   (c) descriptive writing
   (b) purely theoretical
   (d) scope definition

8. Which is NOT a good scope boundary?

   (a) dataset
   (c) tools
   (b) timeframe
   (d) "everything"

9. Interpretation primarily asks:

   (a) What does the result mean?
   (c) Who printed the report?
   (b) What font was used?
   (d) What is the page number?

10. Calibration and repeatability are especially important for:

(a) research instrumentation

(b) research titles

(c) roll numbers

(d) acknowledgements

Rapid-fire: One-word / one-line

1. Define research in your own words:
   Research is a systematic, scientific investigation to find new facts, verify knowledge and answer questions using evidence.

2. One difference between a topic and a research problem:
   A topic is a broad area (e.g. cybersecurity); a research problem is a specific, researchable question with a gap, scope and measurable outcome.

3. One source of a research problem:
   Literature review / research gaps (also industry problems, emerging trends, patents).

4. One common selection error:
   Choosing a problem that is too broad or vague (also: not feasible, not relevant, ignoring prior art).

5. Expand SMART:
   Specific, Measurable, Achievable, Relevant, Time-bound.

6. One primary-data example in CS:
   Usage logs or CPU readings collected in your own experiment; interview responses; sensor readings.

7. One secondary-data example in CS:
   A Kaggle or GitHub dataset (also IEEE articles, government data).

8. One data collection tool:
   Google Forms for questionnaires (also Wireshark, PerfMon, BeautifulSoup, Selenium).

9. One research instrument:
   Simulators, sensors or benchmarking suites, e.g. the Mininet or NS-3 network simulator.

10. Difference between analysis and interpretation:
    Analysis finds the patterns and measurements in the data; interpretation explains what they mean for the research problem.

14 Part N — Connect the Dots

Match Column A with Column B.

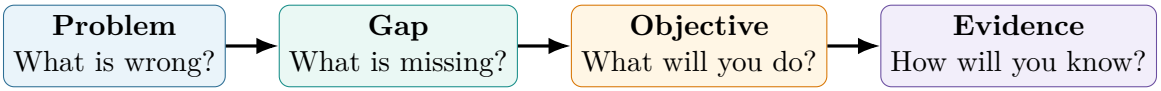
A	Concept	B	Meaning / Example
1	Exploratory	A	Collected directly
2	Primary data	B	Clear, measurable, realistic, relevant, time-bound
3	Scope	C	Discover patterns / new ideas
4	SMART	D	Limits and boundaries of a study
5	Empirical	E	Observation / experiment based
6	Instrumentation	F	Tools/hardware/software for collection and analysis
7	Secondary data	G	Existing datasets/articles/logs
8	Interpretation	H	Explain what findings mean for the problem

Answers: 1-C, 2-A, 3-D, 4-B, 5-E, 6-F, 7-G, 8-H.

15 Part O — Viva / Oral Defense Practice

Imagine your teacher asks: *"Defend your research problem in 60 seconds."*

Use the structure:



Your 60-second defense notes:
Problem: Gap: Objective: Evidence/data:

Problem: IoT networks face new (zero-day) attacks that signature-based IDS cannot detect, causing data loss and downtime.

Gap: Existing ML-based IDS are mostly tested on known attacks and are often too heavy for low-resource IoT devices.

Objective: Within 12 weeks, train and compare Random Forest and CNN models on UNSW-NB15 to reach at least 95% accuracy with under 3% false alarms.

Evidence/data: UNSW-NB15 plus lab traffic captured with Wireshark; metrics are accuracy, recall, F1 and false-alarm rate, compared with a baseline.

16 Part P — Unit I Master Check

Can I do it without looking at the PPT?

Rate yourself honestly: **1 = not yet, 2 = partly, 3 = confidently**.

Skill	1	2	3
Explain what research is and why it matters			
Identify the key elements of research			
Generate sources of research problems			
Distinguish a topic from a research problem			
Evaluate a research problem using quality criteria			
Identify common selection errors			
Define scope and boundaries			
Write SMART objectives			
Choose an appropriate research approach			
Distinguish theoretical and empirical research			
Classify primary/secondary data			
Classify qualitative/quantitative data			
Select a suitable collection tool			
Explain analysis vs interpretation			
Select suitable research instrumentation			
Explain why calibration and repeatability matter			

My one-minute takeaway

The most important idea I learned in Unit I is:
A good research problem is clear, narrow, feasible and measurable, and starts from a gap in the literature rather than a trend.

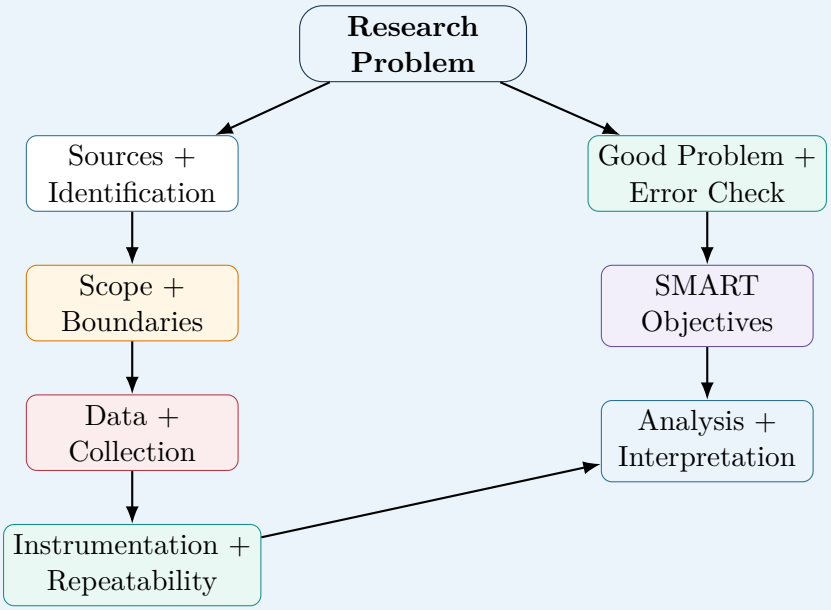
The concept I still need to revise is:
Telling analysis from interpretation, and applying calibration and repeatability in experiments.

One research problem I can now formulate is:
How can a lightweight ML model detect unseen attacks in IoT network traffic?

Exit Ticket: Researcher Mindset

Complete the sentence:
"A good researcher does not start by asking which tool to use *; a good researcher first asks* what problem to solve *."*
Explain your answer in one or two lines:
Tools and trends change, but a clearly defined problem, gap and objective give research its direction, scope and a way to measure success.

Unit I in one visual



End of Unit I Worksheet — Introduction to Research Problem